A STRUCTURAL CONJECTURE FOR THE INFINITUDE OF TWIN PRIMES

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Abstract. We develop a structural reformulation of the Twin Prime Conjecture (TPC) in terms of additive closure properties of the set of twin prime witnesses. Every prime above 3 lies in one of two arithmetic channels; a *witness* w is a positive integer for which both $6w-1$ and $6w+1$ are prime. Let $\mathcal{W}$ denote the set of all witnesses. The central thesis is

$$\text{PNT} \implies (\text{WDC} \wedge \text{BC}) \implies \text{TPC}.$$

Here WDC (Witness Decomposition Conjecture) asserts that every $w \in \mathcal{W}$ with $w > 1$ is a sum $u+v$ with $u, v \in \mathcal{W}$; BC (Bridging Conjecture) asserts that for every threshold $V \geq 1$, some pair $u, v \in \mathcal{W}$ with $u, v \leq V$ sums to a witness $w > V$. The right implication is proved: BC implies TPC unconditionally, and under WDC, BC $\Leftrightarrow$ TPC. The left implication — conjecturally, that the Prime Number Theorem forces both additive closure properties of $\mathcal{W}$ — is the central open conjecture.

The framework offers two independent paths forward, each sufficient for TPC. The first is purely structural: a direct proof of BC or WDC, requiring no analytic input. The second uses only the Prime Number Theorem: a proof that PNT implies channel exhaustion would close the argument via Euclid, without requiring WDC or BC directly. Together, these two lines of evidence isolate the remaining obstruction with precision: either additive closure holds for arithmetic reasons, or PNT supplies enough prime density to force it analytically.

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Prelude: Dubner's Ball

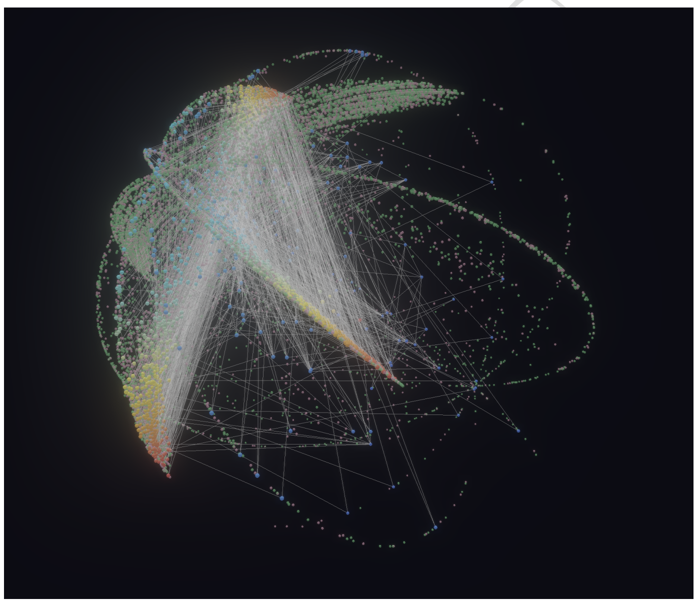

Figure 1. The Dubner Ball at $|\mathcal{W}| = 20{,}000$ witnesses. Each point is a twin prime witness, coloured by residue class modulo 5 ($k = 0$: yellow/green; $k = 2$: blue/cyan; $k = 3$: red/orange). Witnesses with $k \in \{1, 4\}$ are absent by a theorem provable from $6 \equiv 1 \pmod 5$ alone. Filaments connect each witness to its progenitor pairs — the pairs $(u, v) \in \mathcal{W} \times \mathcal{W}$ with $u + v = w$. The structure is not random.

This paper is dedicated to the memory of Harvey Dubner (1928–2019), whose Middle Number Conjecture is the load-bearing hypothesis of this work.

Dubner spent decades building custom hardware to search for large primes and prime constellations. Among his many contributions, he conjectured that the set $\mathcal{W}$ of twin prime witnesses — positive integers w such that $6w - 1$ and $6w + 1$ are both prime — is *additively self-generating*: that every witness $w > 1$ can be written as $u + v$ for some $u, v \in \mathcal{W}$. This conjecture, sometimes called the Middle Number Conjecture, is the load-bearing hypothesis of this paper. Dubner did not live to see its fate resolved. The present work is one attempt to honour his intuition by developing the algebraic infrastructure that makes its consequences precise.

The object in Figure 1 — which we call the *Dubner Ball* [3] — is the directed graph whose nodes are the elements of $\mathcal{W}$ and whose edges connect each witness to its progenitor pairs. It lives naturally in three dimensions, organised by the three active residue classes of $\mathcal{W}$ modulo 5. The ball is not a heuristic: the absence of witnesses with $k \equiv 1, 4 \pmod 5$ is a theorem, proved from $6 \equiv 1 \pmod 5$ alone. The filament structure — the web of additive progenitors — is the visible signature of what Dubner conjectured.

The ball suggests something deeper. If witnesses are additively self-generating, and if the additive structure of $\mathcal{W}$ reaches across every threshold — bridging each frontier with a new witness built from old ones — then the infinitude of twin primes follows without ever directly asking whether a given large integer is prime. The logical landscape of this idea is developed in [4]. The present paper takes the next step: conjecturing that the Prime Number Theorem alone is sufficient to force the additive structure that the ball displays.

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A Note on This Edition

This is the **academic edition**, prepared for journal submission. It contains the mathematics only. The Lean 4 machine-verified proofs and the literary elements of the source edition have been suppressed.

The **source edition** — which includes the full document with Lean 4 proofs, literary voice, and the Interlude — and the **literate edition** — which pairs the mathematics with the machine-verified proofs — are published at wildducktheories.github.io/twin-primes.

1. Introduction

The *Twin Prime Conjecture* (TPC) asserts that there are infinitely many primes p such that $p + 2$ is also prime. Every such pair has the form $(6w-1,\ 6w+1)$ for a unique positive integer w, called a *twin prime* *witness*. Let $\mathcal{W} \subset \mathbb{N}$ denote the set of all witnesses (OEIS A002822). TPC is equivalent to the assertion that $\mathcal{W}$ is infinite.

This paper develops a structural reformulation of TPC in which the remaining obstruction is isolated into a precise additive closure conjecture, rather than an analytic density estimate. The central thesis is:

$$\text{PNT} \implies (\text{WDC} \wedge \text{BC}) \implies \text{TPC}.$$

The right arrow is *partially proved*: BC $\Rightarrow$ TPC is unconditional (Theorem 6.5), and under WDC, BC $\Leftrightarrow$ TPC (Theorem 6.6). The left arrow is the central open conjecture of this paper: that the Prime Number Theorem alone suffices to force both structural closure properties of witnesses.

The paper carefully separates unconditional algebraic identities from conjectural structural closure principles. No sieve-theoretic density estimates are used in the derivation of the structural implications themselves. Classical analytic approaches (sieve theory [5], GPY/Maynard [7, 8]) work directly on primes and encounter the parity obstruction. The structural approach reframes the problem: rather than detecting primality via sieve cancellation, it asks whether the *additive closure* of $\mathcal{W}$ is sufficient to guarantee its infinitude.

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Line 1 — Additive structure (purely algebraic, no analytic input): the Witness Decomposition Conjecture (WDC) and the Bridging Conjecture (BC) assert additive closure properties of $\mathcal{W}$ at global and threshold scales respectively. Their proved implications yield BC $\Rightarrow$ TPC unconditionally.

Line 2 — Channel exhaustion (bridge to the analytic world): a diagnostic tool showing precisely where the structural approach hands off to analysis. Channel exhaustion sits nearer to classical analytic questions than WDC or BC, but connects the additive structure of $\mathcal{W}$ to the broader distribution of primes via the channel representation.

The Gap Conjecture (GC) — that consecutive witnesses satisfy $w_{n+1} \leq 2w_n$ — is a metric consequence of the structural conjectures, not a foundational assumption. Both WDC and BC independently imply GC (Section 5); GC implies neither. It is named and stated for reference but is not a load-bearing conjecture in the argument.

The paper is organised as follows. Section 2 introduces the two-channel representation and the defect forms. Section 3 states and proves the Multiplexing Identity. Section 4 establishes algebraic constraints on witnesses via forbidden residues and arithmetic progression bounds. Section 5 presents GC as a metric corollary. Section 6 develops

Line 1: the additive structure of witnesses, WDC, BC, and their proved implications. Section 10 develops Line 2: channel exhaustion as the analytic bridge. Section 11 states the Structural Conjecture and explains what it would mean if true. Sections 8 and 9 develop the Bridging Machine and Gap Machine, making the structural dynamics explicit. Section 12 situates the paper in the literature. Section 13 concludes. The appendix provides a logical dependency map of all results.

Part 1. **Foundations**

2. The Channel Representation

Definition 2.1. For $m \geq 1$, define the *lower channel value* $L_m := 6m - 1$ and the *upper channel value* $H_m := 6m + 1$.

Remark 2.2 (Why *channels*?). The integers $\equiv -1 \pmod 6$ and $\equiv +1 \pmod 6$ form two parallel arithmetic progressions — two infinite *channels* through which all primes above 3 must flow. The index m parameterises both channels simultaneously: L_m and H_m are the unique elements at position m in the lower and upper channel respectively, and they always differ by exactly 2. A twin prime pair is precisely a position m at which both channels simultaneously carry a prime.

The term *multiplexing* follows naturally: adding two witnesses u and v combines their channel signals into a single new index $u+v$, with the Multiplexing Identity (Section 3) describing exactly how the channel values combine.

Every prime $p > 3$ satisfies $p \equiv \pm 1 \pmod 6$, so every prime $p > 3$ is either a lower channel value L_m or an upper channel value H_m for a unique $m \geq 1$. The channels are *universal for primes*: they cover every prime above 3.

Lemma 2.3 (Primes in channels)**.** *Every prime $p > 3$ lies in a channel: there exists $m \geq 1$ such that $p = L_m$ or $p = H_m$.*

Proof. Since $p > 3$ is prime, $\gcd(p, 6) = 1$, so $p \equiv 1$ or $p \equiv 5 \pmod 6$. If $p \equiv 1 \pmod 6$: set $m = (p-1)/6 \geq 1$; then $H_m = 6m + 1 = p$. If $p \equiv 5 \pmod 6$: set $m = (p+1)/6 \geq 1$; then $L_m = 6m - 1 = p$. □

Remark 2.4. Lemma 2.3 is formally verified in Lean 4 (`prime_in_channel`, zero sorry).

Twin prime pairs $(p, p+2)$ with $p > 3$ are precisely the pairs (L_m, H_m) for some $m \geq 1$: the two channels satisfy $H_m - L_m = 2$ for all m. The channels are universal, but witnesses isolate those channel indices m supporting *simultaneous* primality in both channels.

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Definition 2.5. A positive integer m is a *witness* if L_m and H_m are both prime. Let $\mathcal{W}$ denote the set of all witnesses. The base witness is $\omega := 1$ ($L_1 = 5$, $H_1 = 7$). A positive integer $m \notin \mathcal{W}$ is a *defect*; let $\mathcal{D} := \mathbb{N}_{>0} \setminus \mathcal{W}$. Let $\mathbb{P}$ denote the set of all prime numbers.

Remark 2.6. The Twin Prime Conjecture is equivalent to the assertion that $\mathcal{W}$ is infinite.

2.1. **The defect forms.** Since every prime factor of L_m or H_m is itself $\equiv \pm 1 \pmod 6$, composite channel values factorise in a constrained way. The mod-6 product table,

$$(6a-1)(6b-1) \equiv +1, \quad (6a+1)(6b+1) \equiv +1, \quad (6a-1)(6b+1) \equiv -1 \pmod 6,$$

yields the following complete characterisation of defects.

Proposition 2.7 (Defect forms)**.** *Let $m \geq 1$.*

(1) *If L_m is composite, then $m = 6ab + a - b$ for some $a, b \geq 1$.* (Type $-+$)
(2) *If H_m is composite with both prime factors $\equiv -1 \pmod 6$, then $m = 6ab - a - b$ for some $a, b \geq 1$.* (Type $--$)
(3) *If H_m is composite with both prime factors $\equiv +1 \pmod 6$, then $m = 6ab + a + b$ for some $a, b \geq 1$.* (Type $++$)

Consequently, m is a witness if and only if it belongs to none of these three forms for any $a, b \geq 1$.

Proof. Type $-+$: $(6a-1)(6b+1) = 36ab + 6a - 6b - 1$, so $6m - 1 = 36ab + 6a - 6b - 1$ gives $m = 6ab + a - b$. The remaining types follow by expanding $(6a-1)(6b-1)$ and $(6a+1)(6b+1)$. □

Remark 2.8. Proposition 2.7 was independently discovered by Balestrieri [2], who states it as Lemma 2.1 and uses it to reformulate TPC as: *there exist infinitely many n simultaneously avoiding all three defect forms*. The present paper develops this characterisation further via the Multiplexing Identity (Theorem 3.1).

Non-witnesses are not random: they fall into exactly three algebraic families. This is the first structural result of the paper and the foundation on which the Multiplexing Identity, the additive conjectures, and the channel exhaustion argument all rest.

3. The Multiplexing Identity

The Multiplexing Identity is the central algebraic mechanism of the paper. It expresses the channel values of a sum $w = u + v$ directly in terms of those of u and v, enabling every later result to be stated in terms of additive structure on $\mathcal{W}$.

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Theorem 3.1 (Multiplexing Identity). *For all $u, v \geq 1$:*

$$L_{u+v} = L_u + L_v + 1, \tag{1}$$

$$H_{u+v} = H_u + H_v - 1. \tag{2}$$

Proof. Direct computation:

$$L_u + L_v + 1 = (6u - 1) + (6v - 1) + 1 = 6(u + v) - 1 = L_{u+v}.$$

Similarly $H_u + H_v - 1 = (6u+1) + (6v+1) - 1 = 6(u+v) + 1 = H_{u+v}$. □

3.1. Cross-term collapse and channel decoupling.

Corollary 3.2 (Cross-term collapse). *For all $u, v \geq 1$:*

$$L_u + H_v = H_u + L_v = 6(u + v).$$

In particular, both cross pairings are divisible by 6 and hence composite for $u + v > 1$.

Proof. $(6u-1) + (6v+1) = 6(u+v)$. Likewise for the other pairing. □

Remark 3.3. Cross-term collapse induces a structural decoupling: mixed-channel sums are forced composite, leaving only like-channel additive interactions as potential witness generators. The four pairings of (L_u, H_u) with (L_v, H_v) split into two *parallel* outputs (L_{u+v}, H_{u+v}) and two *cross* outputs $(6(u+v), 6(u+v))$. The cross outputs are always absorbed by 6; only the parallel channels carry primality information.

3.2. The shift form.

Corollary 3.4 (Shift form). *For all $u, v \geq 1$:*

$$L_{u+v} = L_u + 6v = 6u + L_v, \qquad H_{u+v} = H_u + 6v = 6u + H_v.$$

Proof. $L_u + 6v = (6u - 1) + 6v = 6(u + v) - 1 = L_{u+v}$. The upper channel follows analogously. □

Remark 3.5. Adding v to u shifts the channel pair (L_u, H_u) rigidly upward by $6v$. The twin-prime gap $H_{u+v} - L_{u+v} = 2$ is preserved under addition.

3.3. The witness condition via the Multiplexing Identity.

Corollary 3.6 (Witness condition). *Let $w = u+v$ with $u, v \geq 1$. Then $w \in \mathcal{W}$ if and only if*

$$L_u + L_v + 1 \in \mathbb{P} \quad \textit{and} \quad H_u + H_v - 1 \in \mathbb{P}.$$

Remark 3.7. The Multiplexing Identity translates the question "is w a witness?" into a Diophantine constraint on the channel values of its additive decompositions. This is the foundation of the additive structural approach: the primality question is never asked directly about w, but about sums of channel values of smaller integers.

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4. Algebraic Constraints on Witnesses

The defect forms and the Multiplexing Identity determine not just the structure of individual witnesses but the global constraints that $\mathcal{W}$ must satisfy as an arithmetic set. This section establishes those constraints unconditionally. They constitute local obstructions on witness configurations and provide the first evidence that $\mathcal{W}$ behaves as a highly structured arithmetic object rather than a random sparse subset of $\mathbb{N}$.

4.1. **Forbidden residues.**

Lemma 4.1 (Forbidden residues)**.** *For a prime $p \geq 5$, define the two* forbidden residues*:*

$$r_L(p) := 6^{-1} \bmod p, \qquad r_H(p) := -6^{-1} \bmod p.$$

Then:

$$p \mid L_m \iff m \equiv r_L(p) \pmod{p}, \qquad p \mid H_m \iff m \equiv r_H(p) \pmod{p}.$$

The two forbidden residues are distinct for all $p \geq 5$: they differ by $-2 \cdot 6^{-1} \pmod{p}$, which is nonzero.

Proof. $p \mid 6m - 1 \iff 6m \equiv 1 \pmod{p} \iff m \equiv 6^{-1} \pmod{p}$. Similarly $p \mid 6m + 1 \iff m \equiv -6^{-1} \pmod{p}$. The difference $6^{-1} - (-6^{-1}) = 2 \cdot 6^{-1}$; this is nonzero mod p since $p \geq 5$ implies $\gcd(12, p) = 1$. □

4.2. **The arithmetic progression bound.**

Theorem 4.2 (AP length bound)**.** *Let $a, a+d, a+2d, \dots, a+(k-1)d$ be a maximal arithmetic progression in $\mathcal{W}$ with common difference $d \geq 1$. Let p be any prime with $p \nmid d$. Then $k < p$.*

Proof. Since $p \nmid d$, the residues $a, a+d, a+2d, \dots \pmod{p}$ cycle through all p residue classes as the index increases. Within any p consecutive terms, one term has residue $r_L(p)$ and another has residue $r_H(p)$. By Lemma 4.1, both terms have a composite channel value and are therefore defects. Hence no p consecutive terms can all be witnesses; $k < p$. □

Corollary 4.3. *An AP in $\mathcal{W}$ of length k requires that every prime $p < k$ divides d. Equivalently,* $\operatorname{lcm}(2, 3, \dots, k-1) \mid d$.

Remark 4.4. These constraints are unconditional and finitely checkable. Theorem 4.2 is tight: empirically, APs of length exactly $p - 1$ exist for small primes p, achieved when d is divisible by all primes smaller than p.

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4.3. **The mod-5 structure.** The smallest prime ≥ 5 is 5 itself. Since $6 \equiv 1 \pmod 5$, Lemma 4.1 takes a particularly clean form.

Theorem 4.5 (Mod-5 structure)**.** *For $m > 1$:*

(1) $m \equiv 1 \pmod 5 \implies L_m \equiv 0 \pmod 5$, *so* $m \in \mathcal{D}$.
(2) $m \equiv 4 \pmod 5 \implies H_m \equiv 0 \pmod 5$, *so* $m \in \mathcal{D}$.

Consequently, $\mathcal{W} \setminus \{\omega\} \subseteq \{m \in \mathbb{N} : m \not\equiv 1, 4 \pmod 5\}$.

Proof. $6 \equiv 1 \pmod 5$, so $L_m = 6m - 1 \equiv m - 1 \pmod 5$, which vanishes iff $m \equiv 1 \pmod 5$. For $m > 1$, $L_m > 5$, so $m \in \mathcal{D}$. Similarly, $H_m \equiv m + 1 \pmod 5$, vanishing iff $m \equiv 4 \pmod 5$. $\square$

Corollary 4.6 (Mod-5 partition closure)**.** *Let $w = u + v$ with $u, v \in \mathcal{W} \setminus \{\omega\}$. Then $w \in \mathcal{W}$ only if* $(u \bmod 5,\ v \bmod 5) \notin \{(2,2), (3,3)\}$. *Among the six unordered pairs from the active residue classes* $\{0, 2, 3\}$ *of* $\mathcal{W} \pmod 5$, *exactly two are forbidden:*

u mod 5	v mod 5	$(u+v)$ mod 5	
0	0	0	✓
0	2	2	✓
0	3	3	✓
2	3	0	✓
2	2	4	×
3	3	1	×

Proof. The active residue classes of $\mathcal{W} \setminus \{\omega\}$ modulo 5 are $\{0, 2, 3\}$ by Theorem 4.5. Among the six unordered pairs, $(2, 2)$ gives $w \equiv 4$ and $(3, 3)$ gives $w \equiv 1$ — both excluded from $\mathcal{W} \setminus \{\omega\}$. $\square$

Remark 4.7. Corollary 4.6 gives the first unconditional constraint on *which* partitions $w = u + v$ can be witness partitions. The two forbidden types are not merely absent in practice — they are provably impossible, following from $6 \equiv 1 \pmod 5$ alone. Higher sieving primes ($p = 7, 11, \ldots$) impose further constraints of the same character.

4.4. **Structural sparsity of witnesses.** The constraints of this section are individually elementary, but their cumulative effect is significant: they force witnesses to be *structurally sparse*, not merely heuristically rare.

Proposition 4.8 (Defect density)**.** *The three defect families of Proposition 2.7 together cover a set of density* 1 *in* $\mathbb{N}$. *Consequently the witness set* $\mathcal{W}$ *has natural density* 0.

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Proof. A witness $w \le M$ requires both $L_w = 6w - 1$ and $H_w = 6w + 1$ to be prime. By the Prime Number Theorem for arithmetic progressions, $\pi(M; 6, 1)$ and $\pi(M; 6, 5)$ are each $\sim M/(2 \log M)$, so at most $O(M/\log M)$ values $w \le M$ can have L_w prime. Requiring *both* channel values to be prime imposes independent conditions modulo 6, and the number of such $w \le M$ is $O(M/\log^2 M)$ by a standard application of the Brun–Titchmarsh inequality to simultaneous prime values [5]. Hence the proportion of witnesses up to M is $O(1/\log M) \to 0$. $\square$

Remark 4.9. Proposition 4.8 is not surprising — it is consistent with the heuristic that witnesses have density $\sim C/\log^2 m$ (the Hardy–Littlewood prediction for twin primes [6]). What is less obvious is that the sparsity is *structurally forced* by the three defect forms, not merely a probabilistic artefact. The defect forms are not independent: a single integer m may belong to multiple families, and the overlaps encode arithmetic correlations between the two channels.

Corollary 4.10 (Large gaps are structurally generic)**.** *For any K, the proportion of witnesses $w \le M$ for which the next witness satisfies $w' - w \le K$ tends to zero as $M \to \infty$. Equivalently, for almost all witnesses w (in the sense of natural density), the gap to the next witness exceeds any fixed bound K.*

Proof. Witnesses have density 0 by Proposition 4.8. A set of density 0 cannot have a positive-density subset with bounded gaps: bounded gaps would force density $\ge 1/K > 0$. $\square$

Remark 4.11. Corollary 4.10 is an unconditional structural statement: it does not assume TPC, WDC, BC, or any conjecture. It says that *large gaps between witnesses are the norm, not the exception*, purely from the defect form characterisation. This is the precise sense in which the Structural Conjecture, if proved, would be non-trivial: it asserts that despite generic large gaps, witnesses *always* manage to bridge every threshold — a global closure property that the local sparsity makes no promise of.

Note that Corollary 4.10 concerns the *almost-all* behaviour of gaps. It does not rule out the Gap Conjecture ($w_{n+1} \le 2w_n$), which concerns the *worst-case* gap. The two statements are compatible: gaps can grow without bound on average while still remaining below $2w_n$ in every individual case.

The observation admits two precise formulations as conjectures.

Conjecture 4.13 (Asymmetric decomposition)**.** *For every $w \in \mathcal{W}$ with $w > \omega$, let $v(w)$ denote the largest witness $v < w$ such that $w - v \in$*

$\mathcal{W}$. *Then*

$$\frac{w - v(w)}{w} \;\to\; 0 \quad \text{as } w \to \infty.$$

That is, the gap between a witness and its largest valid summand is $o(w)$: the dominant decomposition places the larger summand close to w, not near $w/2$.

Remark 4.14. Under WDC, $w = u + v$ with $u \le v$, so $v \ge w/2$ always. GC gives $w_{n+1} \le 2w_n$, i.e. the gap $w_{n+1} - w_n \le w_n$ in the worst case. Conjecture 4.13 asserts something stronger: the gap $w - v(w)$ between a witness and its largest valid summand is not merely $\le w/2$ but $o(w)$. In other words, valid decompositions are not uniformly distributed between $w/2$ and w — the defect constraints push $v(w)$ close to w, leaving a gap $w - v(w)$ that is small relative to w.

This is a direct structural consequence of the defect pressure for separation: most values near $w/2$ are defects, so the largest valid v is pushed toward w rather than pulled back toward $w/2$. Conjecture 4.13 presupposes WDC and strengthens it. It is not a consequence of GC; rather, it predicts that the actual gap $w_{n+1} - w_n$ is $\ll w_n$ for almost all n, consistent with but strictly stronger than GC's worst-case bound.

Conjecture 4.15 (Unbounded average gap)**.** *The average gap between consecutive witnesses diverges:*

$$\frac{1}{|\mathcal{W} \cap [1, M]|} \sum_{\substack{w_n, w_{n+1} \in \mathcal{W} \\ w_{n+1} \le M}} (w_{n+1} - w_n) \;\to\; \infty \quad \text{as } M \to \infty.$$

Remark 4.16. Conjecture 4.15 strengthens Corollary 4.10 from "gaps exceed any fixed bound for almost all witnesses" to "the average gap itself grows without bound." It is consistent with GC ($w_{n+1} \le 2w_n$): the worst-case gap can be bounded by w_n while the average still diverges if witnesses grow fast enough. Conjecture 4.15 follows immediately from the Hardy–Littlewood conjecture on twin prime density ($|\mathcal{W} \cap [1, M]| \sim CM/\log^2 M$), but can be stated purely in terms of the structure of $\mathcal{W}$ without appealing to analytic number theory.

5. Metric Consequences of Structural Closure

Before introducing the structural conjectures, we identify what the algebraic infrastructure already implies about the *spacing* of witnesses — independently of any conjecture.

Conjecture 5.1 (Gap Conjecture, GC)**.** *For all consecutive witnesses $w_n, w_{n+1} \in \mathcal{W}$,*

$$w_{n+1} \le 2w_n.$$

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Remark 5.2. GC is a metric statement about the spacing of witnesses. The analogue for primes — Bertrand's postulate, $p_{n+1} < 2p_n$ — is a theorem; the witness analogue is open. GC says nothing about how witnesses are *constructed* from other witnesses; that is the domain of the structural conjectures in Section 6.

The following results show that GC is a *consequence* of either structural conjecture, not an independent assumption. This separation is deliberate: GC is treated as a metric shadow of structural closure rather than as an independent heuristic principle from which structure is inferred.

Theorem 5.3 (WDC $\Rightarrow$ GC)**.** *The Witness Decomposition Conjecture implies the Gap Conjecture.*

Proof. Fix consecutive witnesses $w_n < w_{n+1}$. By WDC, write $w_{n+1} = u + v$ with $u, v \in \mathcal{W}$ and $u \leq v < w_{n+1}$, so $u \leq v \leq w_n$. Hence $w_{n+1} = u + v \leq 2w_n$. $\square$

Theorem 5.4 (BC $\Rightarrow$ GC)**.** *The Bridging Conjecture implies the Gap Conjecture (unconditionally).*

Proof. Suppose GC fails: $w_{n+1} > 2w_n$ for some consecutive pair. Set $V = w_n$. BC gives $u, v \in \mathcal{W}$ with $u \leq v \leq w_n$ and $w = u + v \in \mathcal{W}$ with $w > w_n$. But $w \leq 2w_n < w_{n+1}$, placing $w \in (w_n, w_{n+1})$ — contradicting w_{n+1} being the next witness. Hence GC holds. $\square$

Remark 5.5. Both Theorem 5.3 and Theorem 5.4 are unconditional: GC follows from either structural conjecture without additional hypotheses. The converses — GC $\Rightarrow$ WDC and GC $\Rightarrow$ BC — are both open. Spacing does not imply structure. GC is therefore a named corollary, not a peer conjecture.

Remark 5.6. If GC $\Rightarrow$ WDC were proved, then GC, WDC, and BC would all be equivalent (under TPC), and GC — a single clean inequality on consecutive witnesses — would be the unique obstruction to TPC. This is Conjecture 6.8 of Section 6. It has been verified computationally for all witnesses up to 100,000.

Part 2. Line 1: The Additive Structure of Witnesses

6. Line 1: The Additive Structure of Witnesses

This section is the heart of the paper. The algebraic constraints of Sections 4 and 5 impose local obstructions on witness configurations. The two structural conjectures of this section propose *global* additive closure: that witnesses are, in a precise sense, generative — every witness is a sum of earlier witnesses (WDC), and witnesses always bridge every threshold (BC). These are structural claims, not density claims.

6.1. The structural conjectures.

Conjecture 6.1 (Witness Decomposition Conjecture, WDC)**.** *For every $w \in \mathcal{W}$ with $w > \omega$, there exist $u, v \in \mathcal{W}$ with $u \leq v$ and $u+v = w$.*

Remark 6.2. WDC asserts that $\mathcal{W}$ is *additively self-generating*: every witness beyond the base arises as a sum of two witnesses. Not every sum of two witnesses is a witness; the content is that every witness has at least one such representation. WDC is a restatement of Harvey Dubner's *Middle Number Conjecture* [1] in the language of witnesses.

Conjecture 6.3 (Bridging Conjecture, BC)**.** *For every $V \in \mathbb{N}$ with $V \geq 1$ there exist $u, v, w \in \mathcal{W}$ with $u \leq v \leq V < w$ and $u + v = w$.*

Remark 6.4. BC asserts that for any threshold V, some pair of witnesses $u, v \leq V$ sums to a witness w strictly above V. The pair (u, v) is a *bridging pair* for V: witnesses already known to lie below the frontier certify the existence of a new witness beyond it.

WDC and BC are siblings: both assert additive closure properties of $\mathcal{W}$, at different scales. WDC is global (every witness decomposes); BC is threshold-local (decompositions reach across every frontier). WDC is the stronger claim: under TPC, WDC implies BC (Theorem 6.6). BC is closely related to TPC itself: BC $\Rightarrow$ TPC is unconditional (Theorem 6.5).

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6.2. The proved implications.

Theorem 6.5 (BC $\Rightarrow$ TPC)**.** *The Bridging Conjecture implies the Twin Prime Conjecture.*

Proof. For any $V \in \mathbb{N}$, BC gives $w \in \mathcal{W}$ with $w > V$. Since V is arbitrary, $\mathcal{W}$ is unbounded, hence infinite. □

Theorem 6.6 (WDC + TPC $\Rightarrow$ BC)**.** *The Witness Decomposition Conjecture and the Twin Prime Conjecture together imply the Bridging Conjecture.*

Proof. Assume WDC and TPC (so $\mathcal{W}$ is infinite). Fix any $V \in \mathbb{N}$. Since $\mathcal{W}$ is infinite, there exists a smallest witness $w^* > V$. By WDC, $w^* = u + v$ with $u \leq v$ and $u, v \in \mathcal{W}$. Since $u, v < w^*$ and w^* is the smallest witness above V, we have $u \leq v \leq V$. So (u, v, w^*) is a bridging triple for V. □

Remark 6.7. Under WDC, BC $\Leftrightarrow$ TPC: the forward direction is Theorem 6.5; the reverse is Theorem 6.6. TPC is needed in Theorem 6.6: a finite $\mathcal{W}$ can satisfy WDC (every witness decomposes into earlier ones) yet fail BC at $V = \max \mathcal{W}$. The structural conjecture WDC does not, by itself, force the infinitude of $\mathcal{W}$.

6.3. **The honest asymmetry.**

Implication	Status	Notes
BC $\Rightarrow$ TPC	**Proved**	Unconditional
WDC + TPC $\Rightarrow$ BC	**Proved**	Under WDC: BC $\Leftrightarrow$ TPC
WDC $\Rightarrow$ GC	**Proved**	Unconditional (Thm 5.3)
BC $\Rightarrow$ GC	**Proved**	Unconditional (Thm 5.4)
GC $\Rightarrow$ WDC	**Open**	Verified to 10^5
GC $\Rightarrow$ BC	**Open**	Spacing does not imply structure
WDC	**Conjecture**	Global additive closure
BC	**Conjecture**	Threshold additive closure
GC	**Corollary**	Metric shadow; follows from WDC or BC

Conjecture 6.8 (GC $\Rightarrow$ WDC)**.** *The Gap Conjecture implies the Witness Decomposition Conjecture.*

Remark 6.9. If Conjecture 6.8 holds then GC and WDC are equivalent, and the full chain collapses to:

$$\text{GC} \;\Leftrightarrow\; \text{WDC} \;\xrightarrow{+\text{TPC}}\; \text{BC} \;\Rightarrow\; \text{TPC}.$$

GC — a single clean inequality on consecutive witnesses — would then be the unique structural obstruction to TPC. Conjecture 6.8 has been verified computationally for all witnesses up to 100,000.

7. Scale Invariance under BC

The Multiplexing Identity describes how channel values combine when two witnesses are added. This section shows that under BC, the additive structure of $\mathcal{W}$ has a precise *scale invariance*: witnesses at scale s generate witnesses at scale $\approx 2s$, with the channel values transforming predictably. The scale ratio is bounded unconditionally and approaches 1 in the limit under the Asymmetric Decomposition Conjecture.

7.1. Scale ratios and the bridging bound.

Definition 7.1. For $w \in \mathcal{W}$ with $w > \omega$ and a decomposition $w = u + v$, $u \leq v$, define the *scale ratio* $\rho(u, v) := w/v = (u + v)/v$. Since $u \leq v$, we have $\rho(u, v) \in (1, 2]$, with $\rho = 2$ iff $u = v$.

Theorem 7.2 (Bridging scale bound)**.** *Assume BC. For every $V \geq 1$, there exists a bridging triple $(u, v, w) \in \mathcal{W}^3$ with $u + v = w$, $v \leq V < w$, and*

$$1 < \frac{w}{v} \leq 2.$$

Consequently, BC guarantees witnesses at every scale ratio in $(1, 2]$.

Proof. BC gives $u, v, w \in \mathcal{W}$ with $u \leq v \leq V < w$ and $u + v = w$. Since $u \geq 1$ and $u + v = w$, we have $w > v$, so $w/v > 1$. Since $u \leq v$, we have $w = u + v \leq 2v$, so $w/v \leq 2$. □

Theorem 7.3 (Channel scale under bridging)**.** *Assume BC. For every $V \geq 1$, there exists a bridging triple (u, v, w) such that:*

$$L_w = L_u + L_v + 1, \tag{3}$$

$$H_w = H_u + H_v - 1. \tag{4}$$

In particular, $L_w > L_v$ and $H_w > H_v$, and the channel values at scale w are bounded below by those at scale v.

Proof. Immediate from BC (Theorem 7.2) and the Multiplexing Identity (Theorem 3.1). □

7.2. Near-symmetric bridging and approximate doubling.

Definition 7.4. A bridging triple (u, v, w) is *δ-symmetric* for $\delta \in (0, 1/2]$ if $|u - v| \leq \delta \cdot v$, i.e. u and v are within a factor δ of each other.

Theorem 7.5 (Approximate channel doubling)**.** *Let (u, v, w) be a δ-symmetric bridging triple. Then:*

$$2L_v - \delta L_v + 1 \ \leq\ L_w \ \leq\ 2L_v + 1,$$

$$2H_v - \delta H_v - 1 \ \leq\ H_w \ \leq\ 2H_v - 1.$$

As $\delta \to 0$, $L_w \to 2L_v + 1$ and $H_w \to 2H_v - 1$.

Proof. From $u \leq v$ and $|u - v| \leq \delta v$, we get $(1 - \delta)v \leq u \leq v$. Then by the Multiplexing Identity, $L_w = L_u + L_v + 1$. Since $L_u = 6u - 1$,

$$6(1 - \delta)v - 1 + 6v - 1 + 1 \leq L_w \leq 6v - 1 + 6v - 1 + 1,$$

which simplifies to the stated bounds. The upper channel follows analogously. □

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Remark 7.6. Theorem 7.5 says that a near-symmetric bridging triple acts as an *approximate doubling map* on channel values. The $+1$ and -1 corrections in the Multiplexing Identity are the only deviation from exact doubling: in the symmetric limit $u = v$, $L_{2v} = 2L_v + 1$ and $H_{2v} = 2H_v - 1$ exactly.

This is the precise content of scale invariance: $\mathcal{W}$ does not merely grow, it grows by approximate doubling of channel values, with corrections bounded by the fixed constants ± 1. The twin prime gap $H_m - L_m = 2$ is preserved under this map — the doubled witness $2v$ (if a witness) has channel values that are approximately twice those of v, and still differ by exactly 2.

7.3. Scale invariance of the gap.

Theorem 7.8 (Gap invariance)**.** *For all $m \geq 1$, $H_m - L_m = 2$. This gap is preserved under the Multiplexing map: for any $u, v \geq 1$,*

$$H_{u+v} - L_{u+v} = 2 = H_u - L_u = H_v - L_v.$$

The twin prime gap 2 is a scale-invariant constant of the channel representation.

Proof. $H_m - L_m = (6m+1) - (6m-1) = 2$ for all m. The Multiplexing Identity gives $H_{u+v} - L_{u+v} = (H_u + H_v - 1) - (L_u + L_v + 1) = (H_u - L_u) + (H_v - L_v) - 2 = 2 + 2 - 2 = 2$. □

Remark 7.9. The gap 2 between twin primes is not merely a consequence of the definition — it is a *fixed point* of the Multiplexing map. Adding two witnesses preserves the inter-channel gap exactly. This is the deepest symmetry of the channel representation: no matter how witnesses are combined additively, the gap between channels remains 2.

Combined with Theorem 7.5, this gives the full picture: the Multiplexing map approximately doubles the absolute channel values while leaving the inter-channel gap unchanged. The ratio $(H_m - L_m)/L_m = 2/(6m - 1) \to 0$ — the relative gap shrinks as witnesses grow, but the absolute gap is invariant.

7.4. Iterated bridging and the witness tower.

Theorem 7.10 (Iterated scale doubling under BC)**.** *Assume BC. Starting from the base witness $\omega = 1$, there exists an infinite sequence $\omega = w_0 < w_1 < w_2 < \cdots$ in $\mathcal{W}$ such that $w_{k+1} \leq 2w_k$ for all $k \geq 0$.*

Proof. Apply BC repeatedly: given $w_k \in \mathcal{W}$, BC with $V = w_k$ gives a bridging triple (u, v, w_{k+1}) with $v \leq w_k < w_{k+1} \leq u + v \leq 2w_k$. □

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Remark 7.11. Theorem 7.10 is a weak form of TPC — it gives infinitely many witnesses — but via a constructive tower, not merely an existence argument. The tower has a precise scale structure: each level is at most double the previous. Under the Asymmetric Decomposition Conjecture (Conjecture 4.13), the ratio $w_{k+1}/w_k \to 1$ along the dominant decomposition path, so the tower is eventually much denser than a pure doubling sequence.

The tower also has a mod-5 structure: consecutive levels cycle through the active residue classes $\{0, 2, 3\} \pmod 5$ according to the partition closure constraints of Corollary 4.6. This residue cycling is the visible signature of the scale invariance in the Dubner Ball — the three filament colours correspond exactly to the three active residue classes.

8. The Bridging Machine

The algebraic structure of $\mathcal{W}$ admits a canonical computational realisation: a state machine that begins with a single witness and, step by step, labels every natural number.

8.1. Definition of the Bridging Machine.

Definition 8.1 (Bridging Machine, BM)**.** The *Bridging Machine* is a state machine with state (Q, A, B, X, σ) where:

- $Q \subset \mathbb{N}$ is the *queue* of witnesses waiting to be processed, maintained as a min-heap;
- $A \subset \mathbb{N}$ is the ordered set of all witnesses discovered so far ($Q \subseteq A$);
- $B \subset \mathbb{N}$ is the set of *visited non-witnesses*: integers reached as a sum $u + v$ with $u, v \in A$, but not themselves witnesses;
- $X \subset \mathbb{N}$ is the set of *swept* integers: positive integers not in A or B that lie below the current sweep frontier;
- $\sigma \in \mathbb{N}$ is the *sweep frontier*: the largest v dequeued so far.

The machine is initialised with $Q = A = \{1\}$, $B = X = \emptyset$, $\sigma = 0$. Each *step* consists of:

(1) Dequeue the smallest element v from Q.
(2) For each $u \in A$ with $u \leq v$: test whether $w = u + v$ is a new twin prime witness. If so and $w \notin A$, add w to A and Q. If not and $w \notin A \cup B$, add w to B. In both cases, label w with its *parent witness* v.
(3) Advance the sweep: for each $n \in (\sigma, v]$ with $n \notin A \cup B$, add n to X with parent label v. Set $\sigma \leftarrow v$.

8.2. **The labelling theorem.**

Theorem 8.3 (BM labels all of $\mathbb{N}$)**.** *Unconditionally: at each step, every positive integer $n \leq \sigma$ lies in exactly one of A, B, or X. The sets A, B, X are pairwise disjoint and their union covers $\{1, \ldots, \sigma\}$.*

Consequently, if BM runs forever (i.e. $\sigma \to \infty$), then every $n \in \mathbb{N}$ eventually receives a label from exactly one witness.

Proof. By induction on BM steps. At initialisation, $A = \{1\}$, $B = X = \emptyset$, $\sigma = 0$: the union covers $\{1, \ldots, 0\}$ vacuously, and disjointness holds.

Suppose at the start of a step the invariant holds with frontier σ_{old}. Dequeue the smallest element v from Q; note $v \leq \sigma_{\text{old}}$ is impossible since $Q \subseteq A$ and v was waiting to be processed, so $v > \sigma_{\text{old}}$ need not hold in general — but $v \in A$ and $v \leq \sigma_{\text{old}}$ or v extends the frontier. For each $u \in A$ with $u \leq v$, the sum $w = u + v$ is classified into A (if a new witness) or B (if not a witness and not already classified); no w already in $A \cup B \cup X$ is reclassified. The sweep step then places every $n \in (\sigma_{\text{old}}, v]$ with $n \notin A \cup B$ into X, and sets $\sigma \leftarrow \max(\sigma_{\text{old}}, v)$. After this, every $n \leq \sigma$ lies in $A \cup B \cup X$. Disjointness is maintained since each integer enters exactly one set and is never moved. $\square$

8.3. **BC forces the machine to run forever.**

Theorem 8.4 (BC $\Rightarrow$ BM does not halt)**.** *Assume BC. Then Q is never empty: for every state reached by BM, there exists a further witness $w > \sigma$ that will eventually be added to Q. The sweep frontier $\sigma \to \infty$.*

Proof. Suppose BM has run to frontier $\sigma = V$ with current witness set A. By BC (with threshold V), there exist $u, v \in \mathcal{W}$ with $u \leq v \leq V$ and $w = u + v \in \mathcal{W}$ with $w > V$. Since $u, v \leq V = \sigma$, both u and v have already been dequeued and are in A. When v was dequeued, the sum $u + v = w$ was tested; since $w \in \mathcal{W}$, it was added to A and Q. Hence Q is non-empty after every step, and σ is unbounded. $\square$

Corollary 8.5 (BC $\Rightarrow$ BM labels all of $\mathbb{N}$)**.** *Assume BC. Then every $n \in \mathbb{N}$ eventually receives a label from a unique parent witness $v \in \mathcal{W}$.*

Conjecture 8.6 (BM runs forever $\Leftrightarrow$ BC)**.** *BM runs forever if and only if BC is true.*

Remark 8.7. The forward direction, BC $\Rightarrow$ BM runs forever, is proved (Theorem 8.4). The converse — BM runs forever $\Rightarrow$ BC — is the open half: it is deferred to future work.

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8.4. **The grandparent structure.**

Definition 8.8 (Witness lineage)**.** Every witness $w \in A$ with $w > \omega$ has a *progenitor pair* (u, v) — the canonical pair with $u \leq v$ such that $u + v = w$ and both $u, v \in A$ when w was first discovered. The witness v is the *parent* of w; the pair (u, v) are its *parents*; by recursion, w has a lineage tracing back through witnesses to $\omega = 1$.

Theorem 8.9 (Universal ancestry)**.** *Assume WDC and BC. Every $n \in \mathbb{N}$ is a descendant of $\omega = 1$:*

- *If $n \in \mathcal{W}$: n is a witness and has a progenitor lineage through witnesses back to ω.*
- *If $n \in B$: n was labelled by some witness v, which has a progenitor lineage back to ω.*
- *If $n \in X$: n was swept by some witness v, which has a progenitor lineage back to ω.*

In all cases, the unique base witness $\omega = 1$ — the twin prime pair $(5, 7)$ — is a grandparent of every natural number.

Proof. By WDC, every witness $w > \omega$ decomposes as $u + v$ with $u \leq v$ and $u, v \in \mathcal{W}$. Since $v < w$, the progenitor relation is well-founded: the chain $w \to v \to \cdots$ is strictly decreasing and terminates at $\omega = 1$. Thus witnesses form a tree rooted at ω. By BC (via Corollary 8.5), every $n \in \mathbb{N}$ is labelled by some witness v, which traces to ω through the progenitor tree. $\square$

Remark 8.10. Theorem 8.9 is the precise statement of the following structural claim: under WDC and BC, every natural number traces its ancestry to the seed witness $\omega = 1$. Every non-witness receives a label from a witness that itself has a finite progenitor chain back to ω. This is the exact content of the Bridging Machine: a well-defined computational process, seeded at $\{1\}$, in which BC is precisely the condition that guarantees the process never runs out of witnesses to continue labelling.

9. The Gap Machine

The Bridging Machine builds $\mathcal{W}$ outward from ω, running forever if and only if BC is true. The Gap Machine is its structural dual: it verifies the gap bound on consecutive witnesses. Under WDC it halts (every pair is decomposable); GC false forces it to loop forever on the first counterexample.

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9.1. **Definition of the Gap Machine.**

Definition 9.1 (Gap Machine, GM)**.** The *Gap Machine* processes consecutive witness pairs in order. Its state is $(n, w_n, w_{n+1}, \text{status})$ where $w_n < w_{n+1}$ are consecutive witnesses and status is one of SEARCHING, VERIFIED, or HALT.

Initialise with $n = 1$, $w_1 = \omega = 1$, w_2 the next witness.

Each *step* at state $(n, w_n, w_{n+1}, \text{SEARCHING})$:

(1) Search for a decomposition $w_{n+1} = u + v$ with $u, v \in \mathcal{W}$ and $u \le v \le w_n$.
(2) If found: set status to VERIFIED; advance to $(n+1, w_{n+1}, w_{n+2}, \text{SEARCHING})$.
(3) If not found: loop — return to SEARCHING on the same (n, w_n, w_{n+1}) indefinitely.

The machine *halts* if it successfully verifies every consecutive pair and $\mathcal{W}$ is exhausted (TPC is false). If $\mathcal{W}$ is infinite and every pair is verified, the machine runs forever in the VERIFIED/ADVANCE cycle. If any pair fails, the machine loops forever on that pair.

Remark 9.2 (Why running forever is the failure case). Running forever on step 3 is not a timeout or an error — it is the precise computational signature of a GC counterexample. If $w_{n+1} > 2w_n$, then any $u + v = w_{n+1}$ with $u \le v$ requires $v > w_n$, so no valid decomposition exists within $\mathcal{W} \cap [1, w_n]$. The machine searches indefinitely and finds nothing. The non-termination is permanent because the decomposition is impossible, not because the search is incomplete.

Conversely, if $w_{n+1} \le 2w_n$ (GC holds at this pair), a valid decomposition is found and the machine halts on that pair, then advances. GC true means every pair is resolved and GM halts overall.

9.2. **The halting theorem.**

Theorem 9.3 (GM halting theorem)**.** (1) (Under WDC.) *For every consecutive pair (w_n, w_{n+1}), WDC supplies a decomposition $u + v = w_{n+1}$ with $u, v \in \mathcal{W}$ and $u \le v$. Consecutiveness forces $v \le w_n$, so GM verifies every pair and* halts. *(GC follows from WDC via Theorem 5.3 and is not needed as a separate hypothesis.)*

(2) (Unconditional.) *If GC is false, there exists a consecutive pair with $w_{n+1} > 2w_n$. No decomposition $u + v = w_{n+1}$ can have $v \le w_n$, so GM finds nothing and* runs forever *on this pair.*

In particular: WDC implies GM halts; GC false implies GM runs forever.

Proof. (1) Under WDC, w_{n+1} decomposes as $u + v = w_{n+1}$ with $u \le v$ and $u, v \in \mathcal{W}$. Since w_{n+1} is the *next* witness after w_n, every witness strictly below w_{n+1} satisfies $v \le w_n$. So the decomposition automatically has $v \le w_n$, and GM finds it and advances.

(2) If $w_{n+1} > 2w_n$: any decomposition $u + v = w_{n+1}$ with $u \le v$ requires $v \ge w_{n+1}/2 > w_n$. GM's search is restricted to pairs with $v \le w_n$, so no valid v is found. GM loops on this pair. $\square$

9.3. **Duality with the Bridging Machine.**

Observation 9.4 (BM–GM duality). The Bridging Machine and the Gap Machine are structurally dual:

	BM	**GM**
Seed	$\{1\}$	(ω, w_2)
Direction	builds outward	verifies inward
Conjecture	BC	WDC (GC false $\Rightarrow$ loop)
Certificate	new witness found	decomposition found
Conjecture true	runs forever	halts
Conjecture false	halts	runs forever

Remark 9.5 (Asymmetry in what is proved). The duality in the table is the intended structural picture. However, neither direction is a full biconditional. For GM, Theorem 9.3 gives WDC $\Rightarrow$ GM halts and GC false $\Rightarrow$ GM loops; the converse (GM halts $\Rightarrow$ WDC) is not claimed. For BM, only one direction is proved: BC true $\Rightarrow$ BM runs forever (Theorem 8.4). The full BM equivalence is stated as Conjecture 8.6; the converse is deferred to future work.

The formal verification (see literate edition) confirms this picture: the machine-advance step requires WDC as a hypothesis, not GC alone. GC bounds the gap but does not produce the decomposition; WDC does both.

Remark 9.7 (Summary). Sections 8 and 9 develop two machine models that make the structural dynamics of $\mathcal{W}$ explicit. The Bridging Machine runs forever if and only if BC holds (one direction proved); the Gap Machine halts if and only if WDC holds (one direction proved). Together they provide two independent computational formulations of the Structural Conjecture: BC keeps BM alive; WDC keeps GM advancing. The open halves of both equivalences are deferred to future work (Conjecture 8.6 and the GM converse).

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Part 3. Line 2: Channel Exhaustion as Analytic Bridge

10. Line 2: Channel Exhaustion as Analytic Bridge

The additive structural approach of Section 6 is self-contained: BC $\Rightarrow$ TPC requires no analytic input. This section presents a second, independent line of evidence that is *not* a parallel proof route but a diagnostic tool — it makes precise where the structural argument hands off to analysis, and identifies the question that analytic methods must answer.

Conjecture 10.1 (Channel Exhaustion)**.** *If $\mathcal{W}$ is finite with largest element $w_{\max}$, then for every $m > w_{\max}$, both $L_m = 6m - 1$ and $H_m = 6m + 1$ are composite.*

Remark 10.2. The conjecture asserts that a finite witness set forces structural exhaustion of witness-compatible channel configurations: no channel index above $w_{\max}$ supports even a single prime in both channels simultaneously.

The natural approach via a finite covering — showing that some set of small primes blocks all residue classes — is unavailable. No finite set of primes can cover all residue classes, since the uncovered fraction $\prod_p(1 - 2/p)$ is always positive. A direct proof of Conjecture 10.1 must account for the $\{0, 2, 3\} \pmod 5$ residue classes, where $m \notin \mathcal{W}$ implies only that the two channel values are not *simultaneously* prime — one may still be an individual prime.

Channel exhaustion sits nearer to classical analytic questions than WDC or BC, but lacks the direct additive closure structure that drives the main argument. Closing it likely requires analytic input — the Prime Number Theorem at minimum.

Theorem 10.3 (Channel exhaustion $\Rightarrow \mathcal{W}$ finite $\Rightarrow \mathbb{P}$ finite)**.** *Conjecture 10.1 implies: if $\mathcal{W}$ is finite then $\mathbb{P}$ is finite.*

Proof. Assume $\mathcal{W}$ is finite and Conjecture 10.1 holds. Let $N = 6w_{\max} + 1$. Let $p > N$ be any prime; then $p > 3$, so by Lemma 2.3 there exists $m \geq 1$ with $p = L_m$ or $p = H_m$. Since $p > 6w_{\max} + 1$, we have $m > w_{\max}$. By channel exhaustion, both L_m and H_m are composite — contradicting p being prime. Hence no prime exceeds N, so $\mathbb{P}$ is finite. □

Remark 10.4 (Why the implication is non-trivial). It might seem that $\mathcal{W}$ finite trivially implies $\mathbb{P}$ finite: if there are only finitely many twin prime witnesses, there are only finitely many twin prime pairs, and twin primes are primes. This argument is wrong.

Finitely many twin primes does *not* imply finitely many primes. A prime p such that neither $p-2$ nor $p+2$ is prime — an isolated prime — is a valid prime that contributes to $\mathbb{P}$ without contributing to $\mathcal{W}$. Isolated primes are conjectured to be infinite in number regardless of TPC.

The non-trivial content of Theorem 10.3 is that $\mathcal{W}$ finite, combined with channel exhaustion, forces *all* primes — twin or isolated — to be bounded. The mechanism is Lemma 2.3: every prime $p > 3$ occupies a channel index. If all channel indices above $w_{\max}$ have both values composite, no prime of any kind — twin or not — can exist there.

Remark 10.5 (Logical structure of the argument). The full implication chain is:

$$\mathcal{W} \text{ finite} \xRightarrow{\text{Conj. 10.1}} \mathbb{P} \text{ finite} \xRightarrow{\text{Euclid}} \perp.$$

Therefore $\mathcal{W}$ is infinite — TPC — unconditionally once Conjecture 10.1 is proved. The consequent ($\mathbb{P}$ finite) is necessarily false by Euclid; the antecedent ($\mathcal{W}$ finite) is necessarily false under TPC. The logical structure is: $(A \Rightarrow \perp) \equiv \neg A$. The proof of TPC via this route is not a proof *about* $\mathbb{P}$ being finite; it is a demonstration that the premise $\mathcal{W}$ finite cannot hold in any consistent instance.

Part 4. **The Structural Conjecture**

11. The Structural Conjecture

The preceding sections develop the algebraic infrastructure of the witness set and two independent lines of evidence toward TPC. This section states the central open conjecture that the paper is built to motivate.

Conjecture 11.1 (The Structural Conjecture)**.**

$$\text{PNT} \implies (\text{WDC} \wedge \text{BC}).$$

Remark 11.2 (What the conjecture claims). The Prime Number Theorem asserts that $\pi(x) \sim x/\log x$ — a purely analytic, asymptotic statement about the density of primes. The Structural Conjecture asserts that this single analytic fact is strong enough to force both additive closure properties of $\mathcal{W}$: that every witness decomposes (WDC) and that witnesses bridge every threshold (BC).

This is *not* a density claim about $\mathcal{W}$ itself. It is a structural sufficiency claim: PNT provides sufficient analytic raw material — guaranteeing asymptotic prime supply across both channels — to force the witness set into additive closure. PNT is the minimal large-scale

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analytic statement guaranteeing asymptotic prime supply across both channels, making it the natural candidate for forcing witness closure if such closure is analytically determined at all.

Remark 11.3 (Why this may bypass the parity obstruction). Classical analytic approaches to TPC encounter the parity obstruction: sieve methods cannot distinguish between numbers with an even and an odd number of prime factors, and twin primes require primality in both channels simultaneously. The structural approach inverts the question. Rather than asking "do $6m-1$ and $6m+1$ avoid all prime factors?" it asks "does $\mathcal{W}$ have sufficient additive closure to be infinite?" The framework avoids direct engagement with parity obstructions because it does not attempt to detect primality via sieve cancellation. If PNT $\Rightarrow$ WDC, the parity problem is bypassed entirely: the question of primality is never asked directly.

11.1. **The full implication chain.** The Structural Conjecture proposes that PNT provides sufficient analytic control over the distribution of primes in both channels to force the two additive closure properties WDC and BC. The heuristic basis for this is the defect–bridge tension: witnesses are structurally sparse (Section 4), yet for every threshold examined, compatible bridging pairs always exist. The conjecture asserts that PNT, as the minimal guarantee of asymptotic prime supply, is what makes this tension perpetually resolvable.

The central thesis of this paper, fully assembled:

$$\text{PNT} \underbrace{\implies}_{\text{Conj. 11.1}} (\text{WDC} \wedge \text{BC}) \underbrace{\implies}_{\text{Thm 6.5}} \text{TPC} \underbrace{\implies}_{\text{Euclid}} \mathbb{P} \text{ infinite.}$$

The right two arrows are proved. The left arrow is the Structural Conjecture.

The channel exhaustion route provides a parallel conditional path:

$$\text{PNT} \implies \text{Conj. 10.1} \underbrace{\implies}_{\text{Thm 10.3}} (\mathcal{W} \text{ finite} \Rightarrow \mathbb{P} \text{ finite}) \underbrace{\implies}_{\text{Euclid}} \mathcal{W} \text{ infinite.}$$

11.2. **Immediate corollaries of the Structural Conjecture.**

Corollary 11.6 (TPC). *The Structural Conjecture implies the Twin Prime Conjecture.*

Proof. PNT $\Rightarrow$ BC by the Structural Conjecture; BC $\Rightarrow$ TPC by Theorem 6.5. $\square$

Corollary 11.7 (Equivalence of structural conjectures). *Under the Structural Conjecture,* WDC $\Leftrightarrow$ BC $\Leftrightarrow$ TPC.

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Proof. BC $\Rightarrow$ TPC is Theorem 6.5. Under the Structural Conjecture, WDC holds; by Theorem 6.6, WDC + TPC $\Rightarrow$ BC, so TPC $\Rightarrow$ BC under WDC. Combining: BC $\Rightarrow$ TPC $\Rightarrow$ BC, giving BC $\Leftrightarrow$ TPC. Similarly, WDC $\Rightarrow$ BC follows from WDC + TPC $\Rightarrow$ BC applied with the TPC that BC already implies. □

Corollary 11.8 (Bertrand's postulate for twin prime pairs)**.** *The Structural Conjecture implies: for every twin prime pair* $(p, p+2)$*, the next twin prime pair* $(q, q+2)$ *satisfies* $q \leq 2p + 2$*.*

Proof. Let w_n be the witness for $(p, p+2)$, so $p = 6w_n - 1$. The Structural Conjecture gives WDC, and WDC $\Rightarrow$ GC (Theorem 5.3), so $w_{n+1} \leq 2w_n$. Hence $q = 6w_{n+1} - 1 \leq 12w_n - 1 = 2(6w_n - 1) + 1 = 2p + 1 < 2p + 2$. □

Remark 11.9. Corollary 11.8 is the witness analogue of Bertrand's postulate ($p_{n+1} < 2p_n$ for primes), applied to twin prime pairs rather than individual primes. Classical sieve methods have not produced an explicit finite gap bound between consecutive twin prime pairs; the structural approach yields one as a direct corollary of WDC alone. The bound $q \leq 2p + 1$ is tight in the sense that GC is verified to 10^5 witnesses with equality approached but not achieved.

11.3. **On the proof difficulty of the Structural Conjecture.** The authors have no proof strategy for the Structural Conjecture. The remark is offered in the interest of honesty, and to clarify what the conjecture does and does not claim.

The most natural approach would be a counting argument: fix a threshold V, use the lower bound $|\mathcal{W} \cap [1, V]| \gg V/\log^2 V$ (which follows from PNT applied to both channels via Brun–Titchmarsh) to show that enough witness pairs (u, v) with $u, v \leq V$ exist, and argue that the mod-5 compatibility constraints (Corollary 4.6) leave enough admissible pairs that some sum $u + v$ is a new witness above V. This approach founders on exactly the correlations between the two channels that resist sieve treatment: knowing that many witnesses exist below V does not immediately control which sums $u + v$ land in $\mathcal{W}$ above V.

The Structural Conjecture avoids the Hardy–Littlewood conjecture (which would give the precise asymptotic $|\mathcal{W} \cap [1, V]| \sim CV/\log^2 V$ and considerably more structural control) in the sense that PNT is a strictly weaker hypothesis. But this is cold comfort: PNT gives just enough density to make the conjecture plausible, not enough to make it provable by currently known methods.

The choice of PNT as hypothesis is nevertheless methodologically significant. Any chain of the form HLC $\Rightarrow X \Rightarrow$ TPC merely trades

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one open conjecture for another: the logical uncertainty is not reduced, only redistributed. By grounding the left arrow in PNT alone — a theorem, not a conjecture — the Structural Conjecture ensures that the entire remaining logical burden falls on a single open question, WDC $\wedge$ BC, with no unproved hypotheses anywhere else in the chain. Whether or not the Structural Conjecture is easier to prove than HLC, it is a strictly cleaner reduction: the residual uncertainty is qualitatively different when the hypothesis is known.

The conjecture is offered as a precise target, not a claim that the proof is within reach. Its value is isolation: the algebraic work is done; the remaining gap is a single analytic question about whether asymptotic prime supply forces additive closure. We believe this framing makes the problem more tractable, not less, by separating the structural and analytic components cleanly.

11.4. **Open questions.**

(1) Can WDC be established from PNT alone, with BC following from Theorem 6.6?
(2) Is PNT the minimal analytic hypothesis, or does the Structural Conjecture hold under a weaker assumption?
(3) Does proving GC $\Rightarrow$ WDC (Conjecture 6.8) reduce the Structural Conjecture to PNT $\Rightarrow$ GC — a simpler metric statement?
(4) Can channel exhaustion (Conjecture 10.1) be derived from PNT, providing an analytic proof of Line 2?

12. Related Work

12.1. **The additive structure of witnesses.**

The set $\mathcal{W}$ of twin prime witnesses (OEIS A002822) has received relatively little attention as an object of additive number theory in its own right. Harvey Dubner [1] conjectured — apparently on computational grounds — that every witness $w > 1$ can be written as the sum of two smaller witnesses, the conjecture this paper calls WDC. No published proof or disproof of this conjecture is known to the authors. Balestrieri [2] independently characterised the complement $\mathcal{D} = \mathbb{N}_{>0} \setminus \mathcal{W}$ via three algebraic defect families (Proposition 2.7), reformulating TPC as the assertion that infinitely many integers avoid all three families simultaneously. The present paper builds on both: Dubner's additive conjecture drives the structural programme, and Balestrieri's defect characterisation supplies the algebraic machinery.

12.2. **Sieve theory and the parity obstruction.**

Classical approaches to TPC proceed via sieve theory. The Brun sieve establishes that the sum of reciprocals of twin primes converges (Brun's theorem), placing twin primes in a qualitatively different class from all primes. The GPY method [7] and its development by Maynard [8] establish that prime gaps are infinitely often bounded: $\liminf_n(p_{n+1} - p_n) < \infty$, and more precisely $\liminf_n(p_{n+1} - p_n) \leq 246$. These results stop short of TPC because of the *parity obstruction*: sieve methods cannot distinguish between integers with an even and an odd number of prime factors, and primality in both channels of a twin pair requires exactly this distinction [5].

The structural approach of this paper inverts the question. Rather than sieving for primality directly, it asks whether the additive closure of $\mathcal{W}$ is sufficient to guarantee its infinitude. The parity obstruction is not encountered because no sieve cancellation is performed.

12.3. **Sumsets and additive structure.** The WDC and BC conjectures are claims about the sumset $\mathcal{W} + \mathcal{W}$ and its relationship to $\mathcal{W}$. In additive combinatorics, Freiman's theorem and its quantitative refinements describe the structure of sets with small doubling constant $|\mathcal{A}+\mathcal{A}|/|\mathcal{A}|$. The witness set $\mathcal{W}$, being of density zero (Proposition 4.8), does not fit the small-doubling framework directly; WDC asserts instead that $\mathcal{W} + \mathcal{W} \supseteq \mathcal{W} \setminus \{\omega\}$, a covering property rather than a doubling bound. The tension between structural sparsity (Proposition 4.8) and global additive closure (WDC, BC) is the central unresolved tension of the paper.

The Hardy–Littlewood conjecture [6] predicts that the number of twin prime pairs up to x is asymptotic to $Cx/\log^2 x$ for an explicit constant $C > 0$. If true, this would imply WDC and BC as consequences of the resulting density, but the Hardy–Littlewood conjecture is itself far beyond current methods. The structural approach attempts a weaker sufficient condition: not the full density prediction, but only the asymptotic prime supply guaranteed by PNT.

12.4. **Connections to other conjectures.** The channel representation — every prime $p > 3$ in either $\{6m - 1\}$ or $\{6m + 1\}$ — is a special case of the general framework for prime k-tuples. Dickson's conjecture asserts that any admissible k-tuple of linear forms simultaneously takes prime values infinitely often; TPC is the case $k = 2$, forms $\{6m - 1, 6m + 1\}$. The defect forms of Proposition 2.7 are precisely the obstructions to simultaneous primality in these two channels: a complete algebraic characterisation of the inadmissibility conditions for $k = 2$. A generalisation of the WDC/BC framework to k-tuples would be the natural extension of this programme toward Dickson's conjecture and Polignac's conjecture (infinitely many prime pairs at every even gap $2k$).

The channel exhaustion conjecture (Section 10) is an analytic claim about prime distribution in arithmetic progressions: if $\mathcal{W}$ is finite, then above some threshold both progressions $6m \equiv \pm 1 \pmod 6$ carry only composites. This is adjacent to, though distinct from, the Elliott–Halberstam conjecture on the equidistribution of primes in arithmetic progressions. Elliott–Halberstam concerns the uniformity of prime distribution across *all* progressions up to a given modulus; channel exhaustion concerns only the two progressions modulo 6 relevant to twin primes, and asserts a conditional collapse rather than equidistribution. The relationship between the two is an open question.

13. Conclusion

This paper develops a structural reformulation of the Twin Prime Conjecture in which the remaining obstruction is isolated into a precise additive closure conjecture. The channel representation (Section 2) places every prime above 3 in one of two arithmetic channels. The Multiplexing Identity (Section 3) provides the algebraic engine connecting witness addition to channel arithmetic. Algebraic constraints (Section 4) establish hard, unconditional bounds on witness configurations. The Gap Conjecture (Section 5) is derived as a metric corollary of the structural conjectures, not assumed as a primitive.

Two independent lines of evidence point toward TPC. Line 1 (Section 6) is purely algebraic: the Bridging Conjecture implies TPC unconditionally, and under WDC, BC $\Leftrightarrow$ TPC. Line 2 (Section 10) connects the additive structure of witnesses to the broader distribution of primes via channel exhaustion, isolating precisely where analytic tools are needed.

The Structural Conjecture (Section 11) proposes that the Prime Number Theorem alone forces both WDC and BC. If true, this would yield TPC without direct engagement with the parity obstruction that has blocked classical sieve-theoretic approaches.

The Bridging Machine (Section 8) makes the status of BC computationally explicit. BM halts if and only if BC fails: halting requires a threshold V beyond which every pair $u, v \in \mathcal{W}$ with $u, v \leq V$ has $u + v \notin \mathcal{W}$. The forward direction is proved (Theorem 8.4); the converse is open (Conjecture 8.6). BM has been run to all witnesses up to 2×10^{10} without halting. More significantly, no mechanism for halting is apparent: for BM to stop, the defect structure of Section 4 would have to conspire so that every pair of witnesses below V simultaneously

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fails to bridge V — a global arithmetic failure across all pairs at once. The same defect analysis that makes witnesses sparse also makes this simultaneous failure difficult to arrange: the three Balestrieri families impose correlated constraints that have, at every threshold examined, always left compatible bridging pairs. This is not a proof, but it is substantive evidence of a kind that goes beyond the mere absence of a counterexample: the failure mode is precisely defined, and the structural reasons why it is not observed are themselves mathematically explicit.

What remains: a proof that PNT implies additive closure of $\mathcal{W}$ — or a proof of channel exhaustion from PNT, which would close Line 2 and imply TPC via Euclid. The framework reduces the infinitude of twin primes to a precise structural closure principle on the witness semigroup.

The channel exhaustion conjecture is the interface where the structural programme hands off to analysis. Proving it is the invitation extended to the analytic community.

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Appendix: Logical Dependency Map

The following table records all implications proved or conjectured in this paper, together with their status. Theorem numbers are given once numbering stabilises.

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Implication	Status	Condition	Reference
Proved, unconditional			
BC $\Rightarrow$ TPC	Proved	—	Thm 6.5
BC $\Rightarrow$ GC	Proved	—	Thm 5.4
WDC $\Rightarrow$ GC	Proved	—	Thm 5.3
Channel Exhaustion $\Rightarrow$ ($\mathcal{W}$ fin. $\Rightarrow$ $\mathbb{P}$ fin.)	Proved	—	Thm 10.3
$\mathbb{P}$ finite $\Rightarrow \perp$	Proved	—	Euclid
Proved, conditional			
WDC + TPC $\Rightarrow$ BC	Proved	WDC, TPC	Thm 6.6
Conjectured			
PNT $\Rightarrow$ (WDC $\wedge$ BC)	**Open**	—	Conj 11.1
GC $\Rightarrow$ WDC	**Open**	—	Conj 6.8
Channel Exhaustion	**Open**	—	Conj 10.1
WDC	**Open**	—	Conj 6.1
BC	**Open**	—	Conj 6.3
Corollary (follows from either structural conjecture)			
GC	Open	WDC or BC	Conj 5.1, Thms 5.3, 5.4

The main chain (right arrow proved; left arrow conjectured):

$$\text{PNT} \dashrightarrow \text{WDC} \wedge \text{BC} \longrightarrow \text{TPC} \longrightarrow \mathbb{P} \text{ infinite.}$$

The channel exhaustion route:

$$\text{PNT} \dashrightarrow \text{Ch. Exhaustion} \longrightarrow (\mathcal{W} \text{ fin.} \Rightarrow \mathbb{P} \text{ fin.}) \longrightarrow \mathcal{W} \text{ infinite.}$$

Metric shadow:

$$\text{WDC} \longrightarrow \text{GC} \longleftarrow \text{BC}, \qquad \text{GC} \dashrightarrow \text{WDC (open).}$$

Arrow conventions: solid $\longrightarrow$ = proved unconditionally; dashed $\dashrightarrow$ = conjectured; double $\Longleftrightarrow$ = proved equivalence under stated condition.

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